



Constructing financial network based on *PMFG* and threshold method

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HIGHLIGHTS

- Based on *PMFG* and threshold method, we construct a correlation-based network (*PTN*).
- We apply the *ISOMAP* algorithm to show *PTN* in low-dimensional Euclidean space.
- There is a correspondence between the clusters in the Euclidean space and the communities of *PTN*.
- Dramatic fluctuations in the market led to significant changes in the community structure of *PTN*.

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ABSTRACT

Based on planar maximally filtered graph (*PMFG*) and threshold method, we introduced a correlation-based network named *PMFG*-based threshold network (*PTN*). We studied the community structure of *PTN* and applied *ISOMAP* algorithm to represent *PTN* in low-dimensional Euclidean space. The results show that the community corresponds well to the cluster in the Euclidean space. Further, we studied the dynamics of the community structure and constructed the normalized mutual information (*NMI*) matrix. Based on the real data in the market, we found that the volatility of the market can lead to dramatic changes in the community structure, and the structure is more stable during the financial crisis.

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1. Introduction

There are a large number of time series in financial markets such as price series and trading volume series, which lead to many studies focusing on financial correlation-based networks [1–25]. A common feature of these networks is the construction of networks based on the correlation between time series. At present, the construction methods mainly include minimum spanning tree (*MST*) and *PMFG* [1–13], and threshold methods [14–25]. *MST* and *PMFG* are methods based on graph theory that can be used to extract skeletons from the correlation matrix. In the threshold method, the correlation coefficient matrix is converted to a 0–1 matrix by a threshold and used as an adjacency matrix for the network.

In previous studies, the researchers have revealed that there are rich structures in *PMFG*, such as 4-cliques and community structure [3,9,11–13]. These advantages make *PMFGs* have been widely used in different markets or indices. For example, the authors studied *PMFG* based on the world market index [11], and *PMFG* was used to analyze *PMFGs* in different countries' markets [12,13]. Furthermore, since *PMFG* is constructed based on time series, we can analyze the dynamics of *PMFG* and thus have a deeper understanding of the market, and some authors have studied the dynamics of *PMFG* [4–9,12,13].

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Threshold networks have also been widely used in different markets such as the Chinese market [16], US stock market [15], and Korean market [20], or for constructing indices and so on [15]. However, a problem with threshold networks is how to choose thresholds. When we convert a distance matrix to an adjacency matrix, if a small threshold is selected, there may be some isolated nodes or subnetworks in the network. This leads to some inconvenience, for example, in previous studies, there are isolated nodes or subnetworks in the threshold networks constructed by researchers [19,20,25].

Here, we will combine the threshold method and *PMFG* to construct network, for convenience, the *PMFG*-based threshold network is abbreviated as *PTN* in this paper. The core idea is to use *PMFG* to extract a skeleton from the correlation coefficient matrix, and then combine it with the threshold network to generate a network without isolated nodes or isolated sub-networks. *PMFG* is used to ensure that the resulting network is connected and has a rich structure, while the threshold network guarantees that some important edges are preserved, whereas in *PMFG*, these edges may be removed.

In this article, we will study the community structure of *PTN*. In the language of complex network theory, a community is a sub-network in which the members are more closely connected with each other. Community analysis of a complex network provides us a way to explore network construction to find each natural sub-network within the entire network. Many algorithms are proposed for detecting networks in networks such as [26–34]. Here, we will only use one of the classic algorithms to show our method [26]. Furthermore, nodes in the network may be shared by different communities, so some researchers have proposed various algorithms to detect overlapping community structure [35–37]. More details can be found in some related review papers [38–40]. In addition, some indicators and methods are proposed for characterizing community structure or evaluating community detection algorithms [41–46]. In this article, we will apply the normalized mutual information measure (*NMI*) to describe the similarity between the different partitions [44,45].

In order to visualize the communities of *PTN*, we will use *PTN* and manifold learning algorithm to reconstruct the market in low dimensional space. In the algorithm, the data is assumed to lie on an embedded non-linear manifold, which are embedded in a high-dimensional space. At present, many manifold learning algorithms have been proposed for dimensionality reduction and visualization, such as *ISOMAP* [47], *LEE* [48], *LE* [49], *SDE* [50], *HE* [51], manifold charting [52] and Riemannian Manifold Learning [53]. In our study, the *ISOMPA* algorithm is used [47], which is a classical non-linear dimensionality reduction algorithm.

Compared with the threshold network, *PTN* has some advantages. Firstly, there is no isolated node or isolated sub-graphs in the *PTN*, which makes its structure richer than the threshold network. Secondly, based on the manifold learning algorithm, the community structure of *PTN* can be clearly displayed in low dimensional space. Finally, we found that the community structure of *PTN* can be well applied to cluster analysis.

This paper is organized as follows. First, we describe the data, and then give the algorithm used in this paper. In the results section, we construct *PTN* and combine it with *ISOMAP* algorithm for data dimensionality reduction. Finally, we study the *PTN*'s community structure and its application.

2. Data and method

2.1. Data

The daily closing price data of the S&P100 constituent stocks were used, from 2005/1/3–2015/4/24, a total of 2595 days. The stocks with missing data were removed and 93 stocks were selected.

In addition, we also used data from the Chinese market. We selected 162 stocks with no missing data from the constituents of the CSI300 index. From 2013/1/4 to 2015/12/23, a total of 721 trading days.

Before constructing the network, we need to preprocess the price data. First, the logarithmic return series $\{r_i(t)\}$ of stock i is computed, where $r_i(t) = \log(p_i(t+1)) - \log(p_i(t))$, the $\{p_i(t)\}$ is the price series of stock i . Then we compute the Pearson correlation coefficient between the stock i and j , as shown in Eq. (1), where $\langle \rangle$ means to calculate the average. In addition, we will use the distance between stocks later, as shown in Eq. (2). In fact, this distance is equivalent to the Euclidean distance of the normalized return series [54].

$$\rho_{ij} = \frac{\langle r_i r_j \rangle - \langle r_i \rangle \langle r_j \rangle}{\sqrt{(\langle r_i^2 \rangle - \langle r_i \rangle^2)(\langle r_j^2 \rangle - \langle r_j \rangle^2)}} \quad (1)$$

$$d_{ij} = (2 - 2 * \rho_{ij})^{\frac{1}{2}} \quad (2)$$

In the following, the correlation coefficient matrix is denoted by $G = G(i, j)$, where $G(i, j)$ is the correlation coefficient between $\{r_i(t)\}$ and $\{r_j(t)\}$.

2.2. ISOMAP algorithm

In our study, *PTN* will be displayed in low-dimensional Euclidean space through the *ISOMAP* algorithm. The main steps of this algorithm are as follows [47].

Step 1. Define the graph W based on the distance matrix $D = [D(i, j)]$ in the following two ways.

(1) A threshold ϵ is set. If $D(i, j) < \epsilon$, nodes i and j are connected with edge, and edge length is $D(i, j)$ (ϵ – *ISOMAP*).

(2) If i is the k nearest neighbor of j , then i and j are connected, and edge length is $D(i, j)$ ($K - ISOMAP$). We study the threshold network, so the first approach is used, where the threshold selection method is described in Section 2.4.

Step 2. Based on graph W , we compute the shortest distance matrix, denoted as W_s . In this paper, we apply Dijkstra's algorithm to compute the shortest distance matrix [55].

Step 3. Multidimensional scaling analysis is used on the matrix W_s . A operator τ is defined as $\tau(S) = -HSH/2$, where $S(i, j) = W_s(i, j)^2$, $H(i, j) = \delta(i, j) - 1/n$, where the function $\delta(i, j) = 1$ if $i = j$ and 0 otherwise. The eigenvalues of $\tau(S)$ are computed, let $\{\lambda_p\}$ be the set of eigenvalues after descending order. Let v_p^i be the i th component of the p th eigenvector v_p . If we set the dimension is p , then the coordinates of the point i can be expressed as $(\lambda_1^{1/2} v_1^i, \dots, \lambda_p^{1/2} v_p^i)$.

2.3. Community detection algorithm and NMI

Here we use a classic algorithm proposed by Newman [26]. This is an algorithm based on optimal modularity.

Let $W = (V, T)$, where $V = \{v_i\}$ is the node set and $T = [T(i, j)]$ is the adjacency matrix. The degree of the node v_i is $d(i) = \sum_j T(i, j)$. It is assumed that the network W is divided into different communities, we assume that each node v_i is partitioned into a community c_v . Then the modularity Q is defined as Eq. (3), where the $m = \frac{1}{2} \sum_{vw} T(v, w)$ is the number of edges in the network.

$$Q = \frac{1}{2m} \sum_{vw} [T(v, w) - \frac{d(v)d(w)}{2m}] \delta(c_v, c_w) \quad (3)$$

When the modularity Q is greater than 0.3, the network W is considered to have a significant community structure [28].

In order to characterize the similarity between the different partitions, here we use the normalized mutual information (NMI) for analysis [44,45], the calculation steps are as follows. Assuming that the real communities of a network is $C^{real} = \{C_i^{real}, i = 1 \dots, N_{real}\}$, the communities found by the algorithm is $C^{found} = \{C_i^{found}, i = 1 \dots, N_{found}\}$. A confusion matrix $CM = CM(i, j)$ is constructed, where the rows correspond to the real communities, and the columns correspond to the communities that are found. The element $CM(i, j)$ is the number of nodes in the real community C_i^{real} found in the community C_j^{found} . The normalized mutual information (NMI) is defined as Eq. (4), where N is the number of nodes, and $CM_{i..} = \sum_j CM(i, j)$, $CM_{..j} = \sum_i CM(i, j)$.

$$I(C^{found}, C^{real}) = \frac{-2 \sum_{i=1}^{N_{found}} \sum_{j=1}^{N_{real}} CM(i, j) \log(CM(i, j)N / CM_{i..} CM_{..j})}{\sum_{i=1}^{N_{found}} CM_{i..} \log(CM_{i..} / N) + \sum_{j=1}^{N_{real}} CM_{..j} \log(CM_{..j} / N)} \quad (4)$$

2.4. PMFG-based threshold network

PMFG is a method for studying complex systems [3]. The elements in the system are converted to nodes in PMFG. PMFG can be embedded on the sphere. We assume that there are n elements in the graph. PMFG is an extension of MST, but it contains more links ($3(n-2)$), with only $n-1$ links in the MST. In addition, PMFG contains some 3-cliques and 4-cliques, and these structures are not included in the MST. Details of the structure can be found in the original paper [3].

In the following, we present the method of constructing the network based on PMFG and threshold method. The main steps are as follows.

Step 1. The correlation coefficient matrix G is calculated.

Step 2. Based on G , we compute PMFG to get the network $W_{pmfg} = (V, T)$, where T is the adjacency matrix with edge weights and $T(i, i) = 0$.

Step 3. In the following, we introduce the parameter α , which is defined as in Eq. (5), α is the proportion of edges that we keep in the threshold network. Larger α corresponds to more links. In Eq. (5), t_α is the threshold chosen to construct the threshold network.

$$\alpha = \frac{Card\{G(i, j) > t_\alpha, j > i\}}{Card\{G(i, j), j > i\}} \quad (5)$$

Step 4. Based on α and t_α , we construct the threshold network $W_{t_\alpha} = (V, T_{t_\alpha})$, where T_{t_α} is the adjacency matrix with edge weights and $T_{t_\alpha}(i, i) = 0$. The non-diagonal elements of T_{t_α} are shown in Eq. (6).

$$T_{t_\alpha}(i, j) = 0, G(i, j) \leq t_\alpha \quad (6a)$$

$$T_{t_\alpha}(i, j) = G(i, j), G(i, j) > t_\alpha \quad (6b)$$

Step 5. Finally, based on the networks W_{pmfg} and W_{t_α} , a new network $W = (V, T_{ptn})$ is generated by the combination of these two networks. The elements of the weighted adjacency matrix T_{ptn} are obtained as follows: $T_{ptn}(i, j) = G(i, j)$ if at least one of $T_{pmfg}(i, j)$ and $T_{t_\alpha}(i, j)$ is not 0, otherwise $T_{ptn}(i, j) = 0$.

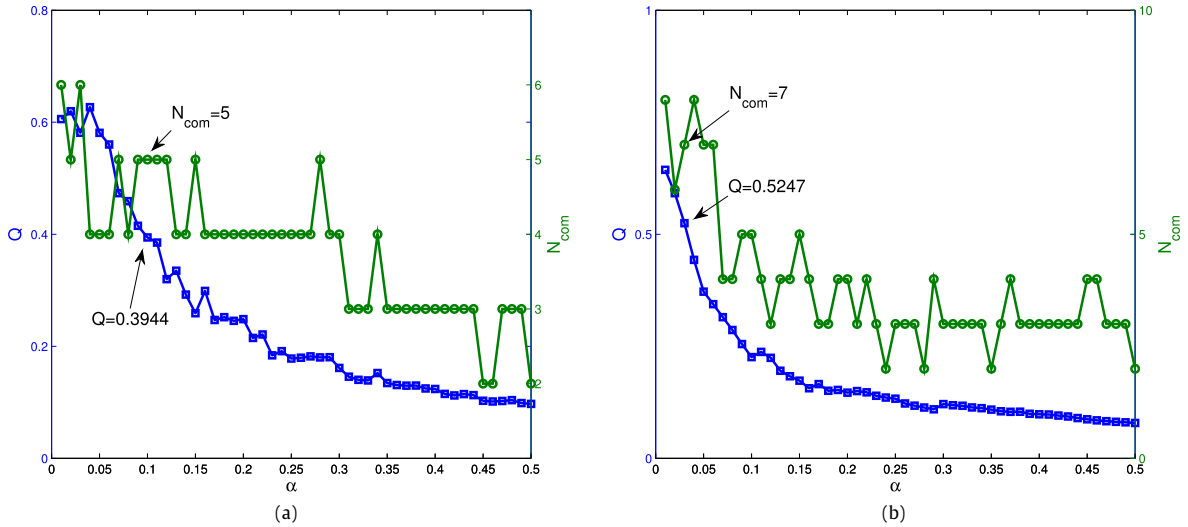


Fig. 1. The figure shows the relationship between α and Q , where figure (a) corresponds to S&P100 index and figure (b) corresponds to CSI300 index. As the value of α increases, the corresponding Q decreases. In addition, the figure also shows that N_{com} decreases as the α increases.

The number of edges in the PTN must be less than the sum of the number of edges in the W_{t_α} and W_{pmfg} ($\alpha * n(n-1)/2 + 3(n-2)$). If n is a large number, the number of edges in PTN is close to the number of edges of the threshold network W_{t_α} ($\alpha * n(n-1)/2$).

We assume that the order of the correlation coefficient matrix is n , and that the complexity of planarity test is $O(n)$ [56]. Then the algorithm complexity of constructing $PMFG$ is $O(n^3)$ [57]. Since the complexity of generating a threshold network is $O(n)$, the complexity of constructing a PTN is still $O(n^3)$.

In Section 3.2, We will apply the $ISOMAP$ algorithm to demonstrate PTN in low-dimensional Euclidean space. Essentially, we are using $\epsilon - ISOMAP$, but our network contains $PMFG$ as a skeleton. Note that before using the $\epsilon - ISOMAP$ algorithm, we need to convert the nonzero elements in T_{tpn} to the distance through Eq. (2). In addition, we select the threshold based on Eq. (5), and we will present the selection method later.

3. Results

3.1. Parameter selection

We first show how to determine the threshold. In the following, we construct $PTNs$ with the constituents of S&P100 index and CSI300 index. We choose different parameters α , and calculate the corresponding PTN . Finally, we detect the community of each PTN and evaluate the result with modularity Q . In addition, PTN is treated as a network with no edge weight when we detect communities. Fig. 1 shows the relationship between α and modularity (Q). We found that the larger α corresponds to the smaller modularity (Q). In addition, the number of communities (N_{com}) also decreases with the increase of α . Larger α corresponds to more links, which makes the community structure more unclear. Thus, the parameter α should not be chosen so large that the network tends to be fully connected, and that edges with higher weights should be preserved. Since $Q = 0.3$ implies a significant community structure, the condition is that the modularity Q corresponding to the parameter α is greater than 0.3 [28]. Therefore, here we choose $\alpha = 0.1$ ($Q = 0.3944$, $N_{com} = 5$) and 0.03 ($Q = 0.5247$, $N_{com} = 7$) for Fig. 1(a) and (b) respectively.

In the following, we always select $\alpha = 0.1$ when calculating the PTN of the S&P100 index, and always choose $\alpha = 0.03$ when calculating the PTN of the CSI300 index. Fig. 2 shows the threshold network and PTN corresponding to S&P100. Fig. 2(a) shows a threshold network with many isolated nodes, while Fig. 2(b) shows a PTN ($\alpha = 0.1$), a connected network.

Furthermore, the communities are detected using the algorithm provided by Newman [26], a total of 5 communities are detected.

3.2. Market visualization

In the following, we use $ISOMAP$ algorithm to display PTN in low-dimensional Euclidean space. Fig. 3 shows the eigenvalues of $\tau(S)$ in descending order. We found that there are a few dominant eigenvalues, which means that the data is hidden in a low-dimensional manifold.

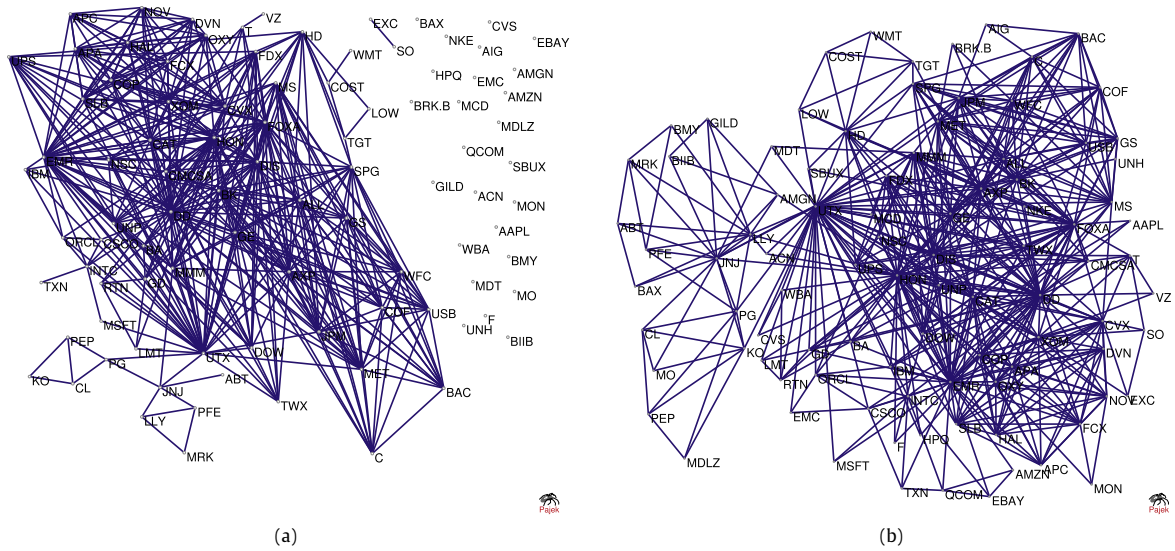


Fig. 2. Figure (a) Shows the threshold network, where $\alpha = 0.1$. (b): A PTN with $\alpha = 0.1$. In the figures, each node corresponds to a company, and letters next to each node are shorthand for company names.

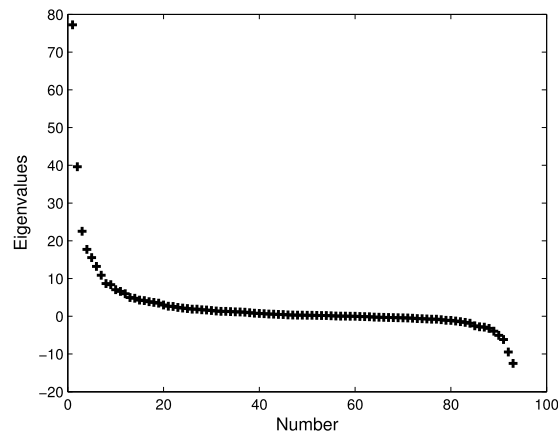


Fig. 3. The descending order of the eigenvalues. From the figure, we can see that there are a few dominant eigenvalues, which means that the data can be reduced to a low-dimensional space.

Fig. 4 shows that PTN is shown in low-dimensional Euclidean space, where figure (a) shows a two-dimensional case and figure (b) shows a three-dimensional case. In Fig. 4, the communities are marked with different symbols, and five communities can also be distinguished intuitively in the low-dimensional space.

The results shown in Fig. 4 imply that there is a correspondence between the vector set in the low-dimensional space and the community. In order to verify this, we use the k-means algorithm to cluster the data after dimension reduction (The k in the k-means algorithm is chosen to be 5.), and calculate the NMI value and Rand index between the clustering result and the community detection result [44,45,58,59]. Fig. 5 shows the results of the comparison, where NMI and Rand index are applied to characterize the similarity between the two partitions. It can be seen that when the dimension is between 2 and 12, most NMI values are greater than 0.68. Similarly, the Rand index is always greater than 0.7, which means that clusters in the low-dimensional space correspond to the communities well.

3.3. Cluster via detecting community structure

We have found that nodes in the same community tend to be clustered together in the low-dimensional space, as shown in Fig. 5. On the other hand, the researchers have provided a method for clustering by detecting communities in the network [60]. In this method, clusters correspond to the communities of the threshold network. Since there are a large number of

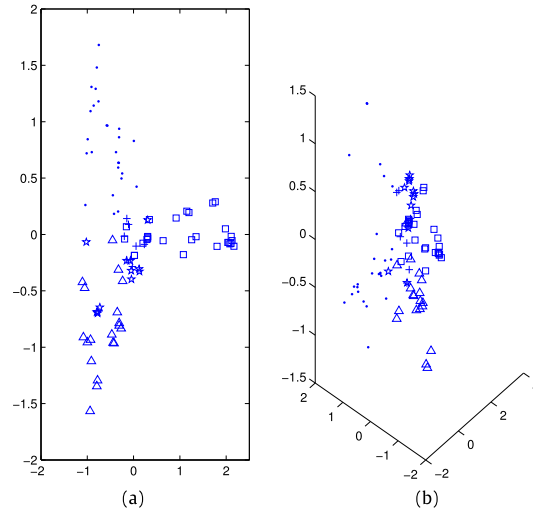


Fig. 4. The points in the five communities are marked with different symbols (point: community 1, +: community 2, triangle: community 3, square : community 4, pentagon : community 5.).

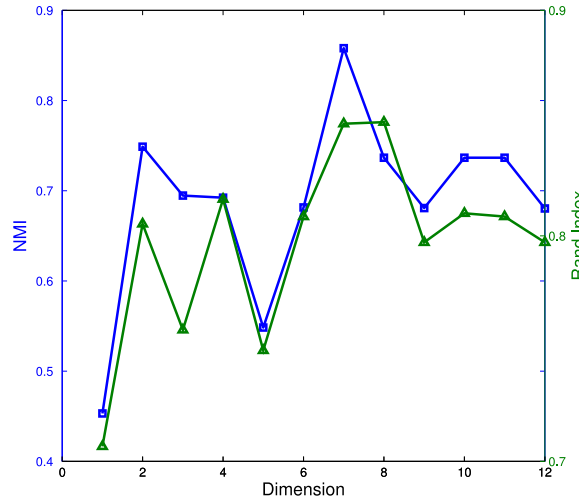


Fig. 5. The *NMI* and *Rand index* are used to measure the difference between the results detected by the community detection algorithm and clusters detected by the clustering algorithm.

isolated nodes in the financial threshold network, as shown in Fig. 2(a), the proposed algorithm by Leonardo N. Ferreira and Liang Zhao cannot be applied directly. Here, we replace the threshold network with *PTN*.

In our study, Fig. 6 shows the adjusted distance matrix in which the nodes in the same community are clustered together as a cluster. In the previous section, we have constructed *PTN* and applied the *ISOMAP* algorithm to show that communities in *PTN* can well correspond to clusters in low-dimensional space. Therefore, our study proposes a modified algorithm that can be used for financial time series.

3.4. Dynamics of community structure

In the previous sections, the results suggested that there is a correspondence between the communities and the clusters in the low-dimensional Euclidean space. On the other hand, since our analysis is based on time series, this means that *PTN* can be considered dynamically. First, one problem is how the community structure in *PTN* changes dynamically. We compute *PTN* with the calculation window for 252 days and the sliding window for 1 day to obtain a series of *PTNs*, denoted as $\{PTN_t\}$. In [13], the authors studied the dynamics of *PMFG*'s community structure in which a series of *PMFGs* and modularity were calculated. Similarly, here we calculate the modularity (Q) of each *PTN*. The results are shown in Fig. 7.

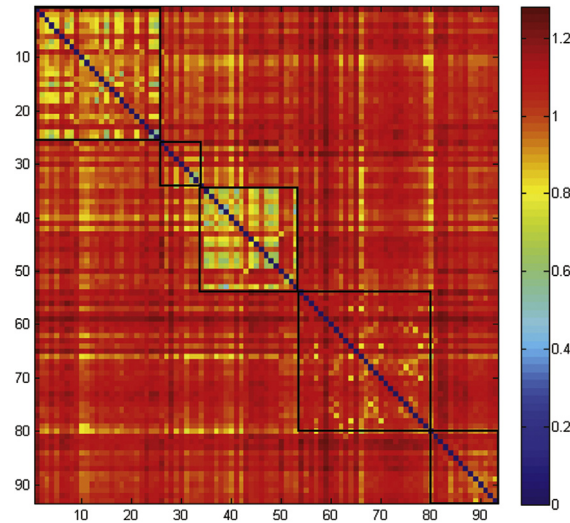


Fig. 6. The figure shows the adjusted distance matrix where each square corresponds to a cluster. The average distance of the five clusters is 0.9640, 0.9633, 0.8930, 1.0595 and 1.0193 respectively. The average distance of the original distance matrix is 1.0503.

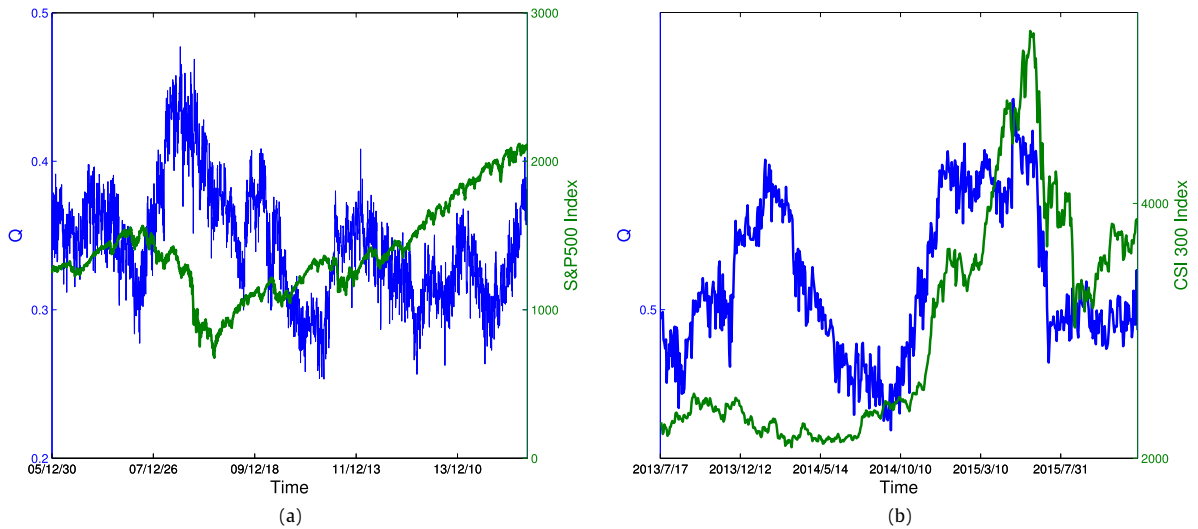


Fig. 7. In figure (a), the modularity (blue line) is compared with the S & P 500 index (green line), where the maximum Q is 0.4772, the minimum is 0.2533, the mean is 0.3451, and the standard deviation is 0.0382. Figure (b) Shows the modularity (blue line) and the CSI300 index (green line), the maximum modularity is 0.7126, the minimum is 0.3783, the mean is 0.5291, and the standard deviation is 0.0762. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In Fig. 7(a), we also show the S&P 500 index. We find that the modularity is significantly larger during the financial crisis, with the maximum being achieved at 2008/7/10 and the corresponding calculation window is 2007/7/8–2008/7/10. In addition, we found that the value of modularity (Q) at different times may vary widely. Similarly, we study the Chinese market, as shown in Fig. 7(b). When the CSI300 index rose to the highest point on 2015/6/8, the CSI300 index is 5353.751, the corresponding modularity is equal to 0.6487. After June 8, the CSI300 index fluctuated sharply and fell rapidly over the next month. Figure (b) shows that the modularity is also significantly reduced. For example, the index is 3663.038 on July 8, and the modularity is equal to 0.5019.

Next, we compare the community detection results with the maximum and minimum modularity, shown in Figs. 8 and 9, where Fig. 8 corresponds to the US market, and Fig. 9 corresponds to the Chinese market. Intuitively, communities with maximum modularity are clearer. These results suggest that financial crises or bear markets have a significant impact on community structure.

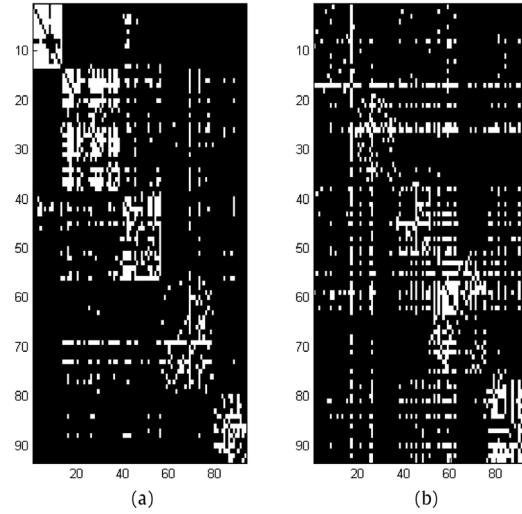


Fig. 8. The adjacency matrix adjusted by the community detection result, the white point corresponds to the edge. Figure (a) and (b), respectively, corresponds to the maximum modularity and minimum modularity. Intuitively, we can see that the figure (a) has a clearer community structure.

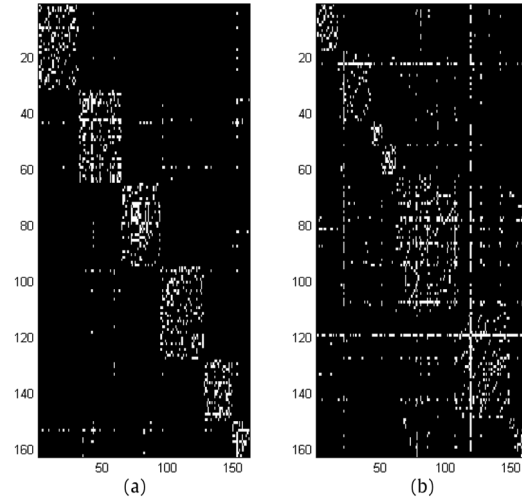


Fig. 9. The adjacency matrix adjusted by the community detection result, the white point corresponds to the edge. Figure (a) and (b), respectively, corresponds to the maximum modularity and minimum modularity. Intuitively, we can see that the figure (a) has a clearer community structure.

Fig. 7 shows that the community structure may change significantly over time, and then, we consider the dynamics of the community. For each PNT_t , we detect its communities, denoted as $V_t = \{V_t(i), i = 1 \dots m_t\}$, where m_t is the number of communities in PNT_t , and then compute the NMI value between V_i and V_j ($i \neq j$). In this way, we obtain an NMI matrix $M_{nmi} = [M_{nmi}(i, j)]$ where $M_{nmi}(i, j) = NMI(V_i, V_j)$ and the result is shown in Fig. 10.

We find that the NMI values change significantly during the initial phase of the financial crisis, and because of the significant change in the NMI values, some rectangular regions are shown in Fig. 10(a). Similarly, the Chinese stock market index declined sharply during the year 2015, and we found that the community structure of the network tends to be more stable as the index declines, resulting in a rectangular area in the NMI matrix (Fig. 10(b)). In addition, Fig. 10(a) and (b) show that the community structure remained stable during the bull market. In short, the calculations show that the switch from the bull market to the bear market corresponds to the dramatic changes in the community structure.

4. Conclusions

In this paper, we apply $PMFG$ and threshold method to construct the correlation-based network (PTN). PTN has no isolated nodes and retains the larger elements in the correlation coefficient matrix, which is different from the threshold network. Although PTN is close to the threshold network when n is sufficiently large, the new network has some particular advantages.

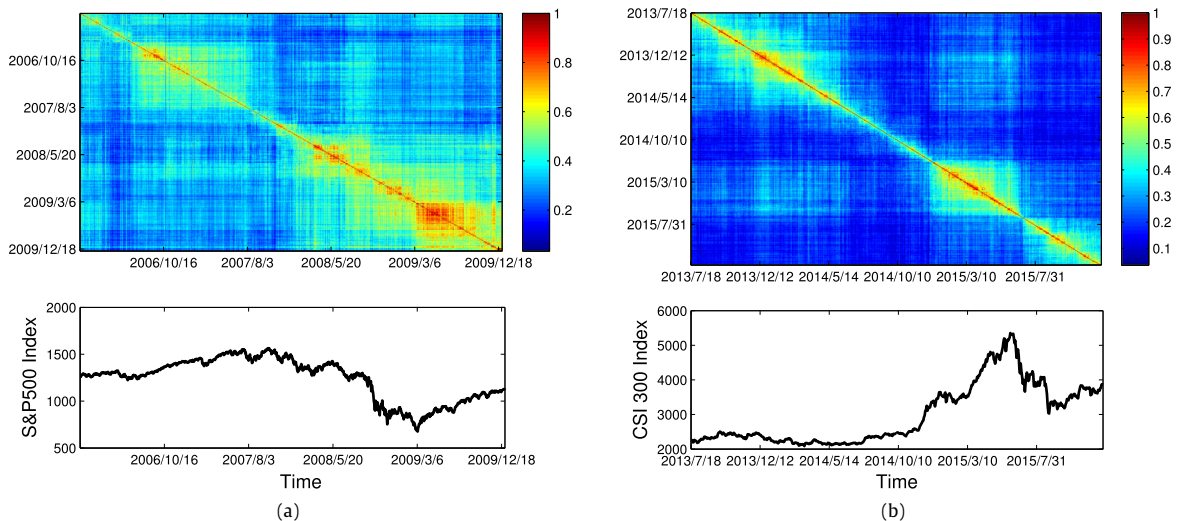


Fig. 10. NMI matrix, the larger the value represents the higher the similarity between the two partitions. It can be seen that during the financial crisis, the similarity is higher, which means that the *PTN* community structure is more stable during this period.

Firstly, based on the *ISOMAP* algorithm, *PTN* can be displayed in low-dimensional Euclidean space. We found that the community structure has a clear geometric meaning. The communities correspond to clusters in low-dimensional space well. Secondly, because there is no isolated node in *PTN*, we study the dynamics of community structure. The results show that the financial crisis or bear market impact on the community structure, resulting in a rapid decline in the modularity. On the other hand, we found that during the financial crisis or bear market, the community structure is more stable, there are some rectangular areas in the *NIM* matrix. Thirdly, in [60], the authors propose a time series clustering algorithm, but there are isolated nodes in the financial threshold network, which makes the original method cannot be directly used for clustering analysis. Our research shows that this algorithm can be modified for financial time series clustering, in which the threshold network is replaced by *PTN*.

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References

- [1] R.N. Mantegna, Hierarchical structure in financial markets, *Eur. Phys. J. B* 11 (1) (1999) 193–197.
- [2] M. Tumminello, T. Matteo, T. Aste, R.N. Mantegna, Correlation based networks of equity returns sampled at different time horizons, *Eur. Phys. J. B* 55 (2007) 209–217.
- [3] M. Tumminello, T. Aste, T. Matteo, R.N. Mantegna, A tool for filtering information in complex systems, *Proc. Natl. Acad. Sci. USA* 102 (30) (2005) 10421–10426.
- [4] F. Pozzi, T. Aste, G. Rotundo, T. Matteo, Dynamical correlations in financial systems, in: D. Abbott, T. Aste, M. Batchelor, R. Dewar, T.D. Matteo, T. Guttman (Eds.), *Microelectronics, MEMS, and Nanotechnology*, International Society for Optics and Photonics, 2007, pp. 68021E–68021E–11.
- [5] D.-M. Song, M. Tumminello, W.-X. Zhou, R.N. Mantegna, Evolution of worldwide stock markets, correlation structure, and correlation-based graphs, *Phys. Rev. E* 84 (2011) 026108.
- [6] M. Eryigit, R. Eryigit, Network structure of cross-correlations among the world market indices, *Physica A* 388 (2009) 3551–3562.
- [7] T. Matteo, F. Pozzi, T. Aste, The use of dynamical networks to detect the hierarchical organization of financial market sectors, *Eur. Phys. J. B* 73 (2010) 3–11.
- [8] T. Aste, W. Shaw, T.D. Matteo, Correlation structure and dynamics in volatile markets, *New J. Phys.* 12 (2010) 085009.
- [9] G. Buccheri, S. Marmi, R.N. Mantegna, Evolution of correlation structure of industrial indices of u. s. equity markets, *Phys. Rev. E* 88 (2013) 012806.
- [10] D.Y. Kenett, M. Tumminello, A. Madi, G. Gur-Gershoren, R.N. Mantegna, E. Ben-Jacob, Dominating clasp of the financial sector revealed by partial correlation analysis of the stock market, *PloS ONE* 5 (12) (2010) e15032.
- [11] I. Vodenska, A.P. Becker, D. Zhou, D.Y. Kenett, H.E. Stanley, S. Havlin, Community analysis of global financial markets, *Risks* 4 (2016) risks4020013.
- [12] X.F. Jiang, T.T. Chen, B. Zheng, Structure of local interactions in complex financial dynamics, *Sci. Rep.* 4 (2014) srep05321.
- [13] L. Zhao, W. Li, X. Cai, Structure and dynamics of stock market in times of crisis, *Phys. Lett. A* 380 (2016) 654–666.
- [14] Y. Yang, H. Yang, Complex network-based time series analysis, *Physica A* 387 (2008) 1381–1386.
- [15] C.K. Tse, J. Liu, F.C. Lau, A network perspective of the stock market, *J. Empir. Finance* 17 (2010) 659–667.
- [16] W.-Q. Huang, X.-T. Zhuang, S. Yao, A network analysis of the chinese stock market, *Physica A* 388 (2009) 2956–2964.
- [17] J.-P. Onnela, A. Chakraborti, K. Kaski, Asset trees and asset graphs in financial markets, *Phys. Scr. T* (106) (2003) 48–54.
- [18] K. Kim, S.Y. Kim, D.-H. Ha, Characteristics of networks in financial markets, *Comput. Phys. Comm.* 177 (2007) 184–185.

- [19] L.S. Junior, Cluster formation and evolution in networks of financial market indices, *Algorithmic Finance* 2 (2013) 3–43.
- [20] A. Nobi, S.E. Maeng, G.G. Ha, J.W. Lee, Effects of global financial crisis on network structure in a local stock market, *Physica A* 407 (2014) 135–143.
- [21] A. Nobi, S. Lee, D.H. Kim, J.W. Lee, Correlation and network topologies in global and local stock indices, *Phys. Lett. A* 378 (2014) 2482–2489.
- [22] V. Boginski, S. Butenko, P.M. Pardalos, Mining market data : A network approach, *Comput. Oper. Res.* 33 (2006) 3171–3184.
- [23] A. Namaki, A.H. Shirazi, R. Raei, G.R. Jafari, Network analysis of a financial market based on genuine correlation and threshold method, *Physica A* 390 (2011) 3835–3841.
- [24] V. Boginski, S. Butenko, P.M. Pardalos, Statistical analysis of financial networks, *Comput. Statist. Data Anal.* 48 (2005) 431–443.
- [25] R.H. Heiberger, Stock network stability in times of crisis, *Physica A* 393 (2014) 376–381.
- [26] M.E.J. Newman, Fast algorithm for detecting community structure in networks, *Phys. Rev. E* 69 (2004) 066133.
- [27] M.E.J. Newman, Modularity and community structure in networks, *Proc. Natl. Acad. Sci.* 103 (23) (2006) 8577–8582.
- [28] A. Clauset, M.E.J. Newman, C. Moore, Finding community structure in very large networks, *Phys. Rev. E* 70 (2004) 066111.
- [29] A. Arenas, A. Fernández, S. Gómez, Analysis of the structure of complex networks at different resolution levels, *New J. Phys.* 10 (2008) 053039.
- [30] H.-J. Li, Z. Bu, A. Li, Z. Liu, Y. Shi, Fast and accurate mining the community structure: Integrating center locating and membership optimization, *IEEE Trans. Knowl. Data Eng.* 28 (9) (2016) 2349–2362.
- [31] P. Ronhovde, Z. Nussinov, Local resolution-limit-free potts model for community detection, *Phys. Rev. E* 81 (2010) 046114.
- [32] J. Reichardt, S. Bornholdt, Statistical mechanics of community detection, *Phys. Rev. E* 74 (2006) 016110.
- [33] H.-J. Li, Y. Wang, L.-Y. Wu, Z.-P. Liu, L. Chen, X.-S. Zhang, Community structure detection based on potts model and network's spectral characterization, *Europhys. Lett.* 97 (2012) 48005.
- [34] A. Lancichinetti, S. Fortunato, Community detection algorithms: A comparative analysis, *Phys. Rev. E* 80 (2009) 056117.
- [35] J. XIE, S. KELLEY, B.K. SZYMANSKI, Overlapping community detection in networks: The state-of-the-art and comparative study, *ACM Comput. Surv.* 45 (4) (2013) 43.
- [36] J. Yang, J. Leskovec, Overlapping community detection at scale : A nonnegative matrix factorization approach, in: *Proceedings of the sixth ACM international conference on Web search and data mining*, pp. 587–596, 2013.
- [37] A. Lancichinetti, S. Fortunato, J. Kertész, Detecting the overlapping and hierarchical community structure in complex networks, *New J. Phys.* 11 (2009) 033015.
- [38] S.E. Schaeffer, Graph clustering, *Comput. Sci. Rev.* 1 (2007) 27–64.
- [39] S. Fortunato, Community detection in graphs, *Phys. Rep.* 486 (2010) 75–174.
- [40] F.D. Malliaros, M. Vazirgiannis, Clustering and community detection in directed networks : A survey, *Phys. Rep.* 533 (2013) 95–142.
- [41] A. Lancichinetti, S. Fortunato, F. Radicchi, Benchmark graphs for testing community detection algorithms, *Phys. Rev. E* 78 (2008) 046110.
- [42] H.-J. Li, H. Wang, L. Chen, Measuring robustness of community structure in complex networks, *Europhys. Lett.* 108 (2014) 68009.
- [43] H.-J. Li, J.J. Daniels, Social significance of community structure: Statistical view, *Phys. Rev. E* 91 (2015) 012801.
- [44] L. Danon, A. Díaz-Guilera, J. Duch, A. Arenas, Comparing community structure identification, *J. Stat. Mech. Theory Exp.* P09008 (2005).
- [45] L. Danon, A. Díaz-Guilera, A. Arenas, The effect of size heterogeneity on community identification in complex networks, *J. Stat. Mech. Theory Exp.* 2006 (11) (2006) P11010.
- [46] D.-M. Song, M. Tumminello, W.-X. Zhou, R.N. Mantegna, Evolution of worldwide stock markets, correlation structure, and correlation-based graphs, *Phys. Rev. E* 84 (2011) 026108.
- [47] J.B. Tenenbaum, V. de Silva, J.C. Langford, A global geometric framework for nonlinear dimensionality reduction, *Science* 290 (2000) 2319–2323.
- [48] S.T. Roweis, L.K. Saul, Nonlinear dimensionality reduction by locally linear embedding, *Science* 290 (2000) 2323–2326.
- [49] M. Belkin, R. Niyogi, Laplacian eigenmaps for dimensionality reduction and data representation, *Neural Comput.* 15 (6) (2003) 1373–1396.
- [50] K.Q. Weinberger, L.K. Saul, Unsupervised learning of image manifolds by semidefinite programming, *Int. J. Comput. Vis.* 70 (1) (2006) 77–90.
- [51] D.L. Donoho, C. Grimes, Hessian eigenmaps: Locally linear embedding techniques for high-dimensional data, *Proc. Natl. Acad. Sci.* 100 (10) (2003) 5591–5596.
- [52] M. Brand, Charting a manifold, *Advances in neural information processing systems* pp. 985–992, 2003.
- [53] T. Lin, H. Zha, Riemannian manifold learning, *IEEE Trans. Pattern Anal. Mach. Intell.* 30 (5) (2008) 796–809.
- [54] R.N. Mantegna, H.E. Stanley, *Introduction to Econophysics: Correlations and Complexity in Finance*, Cambridge University Press, 1999, pp. 105–106.
- [55] E.W. Dijkstra, A note on two problems in connexion with graphs, *Numer. Math.* 1 (1) (1959) 269–271.
- [56] J. Hopcroft, R. Tarjan, Efficient planarity testing, *J. ACM* 21 (4) (1974) 549–568.
- [57] G.P. Massara, T. Matteo, T. Aste, Network filtering for big data: Triangulated maximally filtered graph, *J. Complex Netw.* 5 (2) (2017) 161–178.
- [58] W.M. Rand, Objective criteria for the evaluation of clustering methods, *J. Am. Stat. Assoc.* 66 (336) (1971) 846–850.
- [59] K. Steinhaeuser, N.V. Chawla, Identifying and evaluating community structure in complex networks, *Pattern Recognit. Lett.* 31 (2010) 413–421.
- [60] L.N. Ferreira, L. Zhao, Time series clustering via community detection in networks, *Inform. Sci.* 326 (2016) 227–242.